

**ENERGY MANAGEMENT IN**  
**MICROGRID**  
**GRID CONNECTED MODE**

**DISTRIBUTION AUTOMATION & SMART GRID**

**EC9130**

**13<sup>TH</sup> WEEK PROGRESS TECHNICAL REPORT**



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## Contribution by the members in group

| Tasks                                            | 2022/E/002 | 2022/E/093 | 2022/E/147 |
|--------------------------------------------------|------------|------------|------------|
| <b>1.PV System with boost converter modeling</b> | √          |            | √          |
| <b>2. Simulation using Matlab Simulink</b>       | √          | √          |            |
| <b>3. Progress Report making</b>                 |            | √          | √          |

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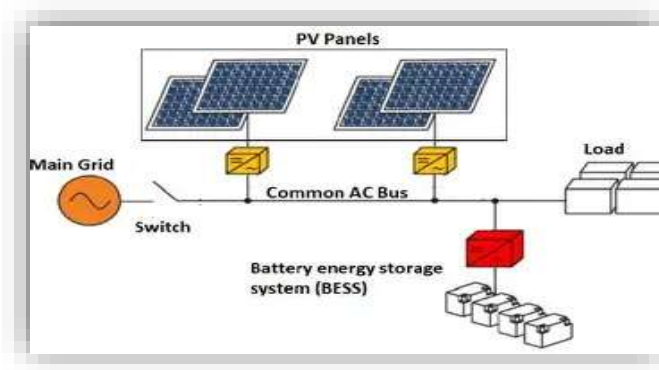
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# **1. INTRODUCTION**

Modern smart grid systems require intelligent coordination between renewable generation, energy storage, and load demand. Due to the intermittent nature of PV generation and dynamic load variations, maintaining power balance and system stability is a critical challenge.

This report presents the progress achieved during Week 13, focusing on the design, modeling, and control of a DC microgrid subsystem, incorporating PV generation, boost converter, Battery Energy Storage System, bidirectional DC-DC converter, and an EMS.

The work corresponds to the system modeling and control implementation phase of the project methodology.



## **2. OBJECTIVES**

### **2.1 Develop PV system with boost converter**

To model the photovoltaic (PV) system and design a boost converter to step up and regulate the PV output voltage, ensuring a stable DC supply despite variations in solar irradiance.

### **2.2 Implement battery energy storage system**

To develop a battery model capable of storing excess energy and supplying power during deficit conditions, while maintaining safe operation through SOC monitoring.

### **2.3 Design bidirectional DC-DC converter**

To enable controlled two-way power flow between the battery and DC bus, allowing efficient charging (buck mode) and discharging (boost mode) operations.

### **2.4 Develop EMS logic**

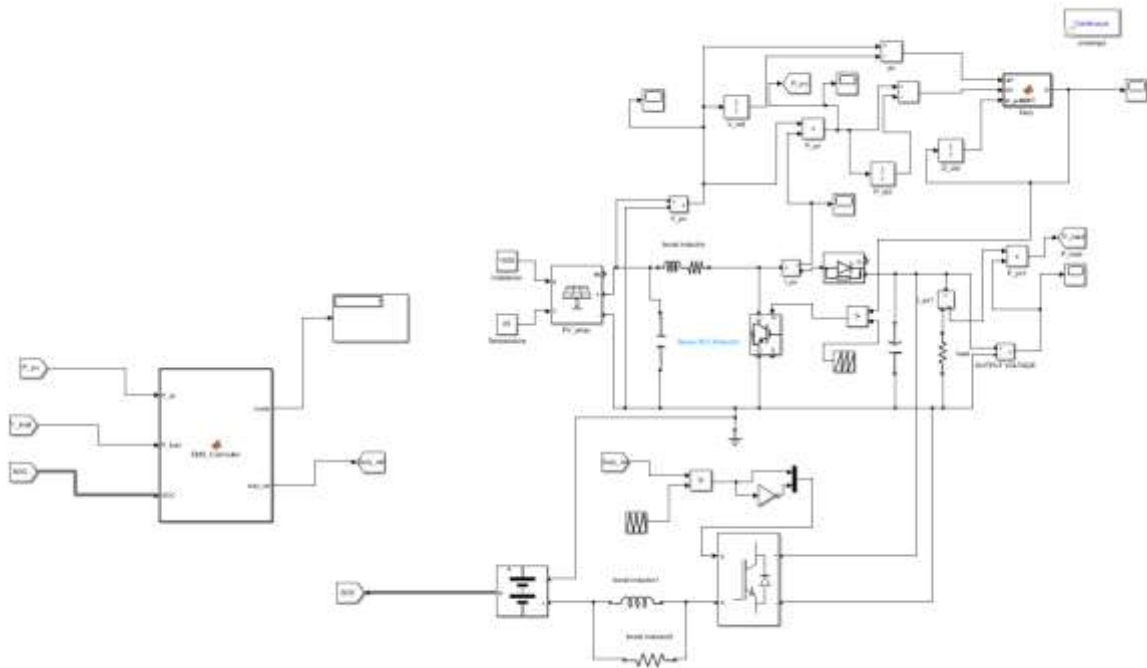
To implement a rule-based control strategy that coordinates power distribution between PV, battery, and load, ensuring optimal energy utilization and system stability.

## **3. SYSTEM CONFIGURATION**

### **3.1 Overall Architecture**

All energy sources and storage devices are interconnected through a common DC bus. The system consists of

- PV array (primary generation source)
- DC-DC boost converter (voltage regulation)
- Battery Energy Storage System
- Bidirectional DC-DC converter
- DC load
- EMS controller



### **3.2 PV System with Boost Converter**

- ✓ The PV array is modeled based on its nonlinear I–V characteristics under
  - Irradiance = 1000 W/m<sup>2</sup>
  - Temperature = 25°C

The PV output power depends on environmental conditions, making it intermittent and variable.

## ✓ Boost Converter Function

The boost converter is used to

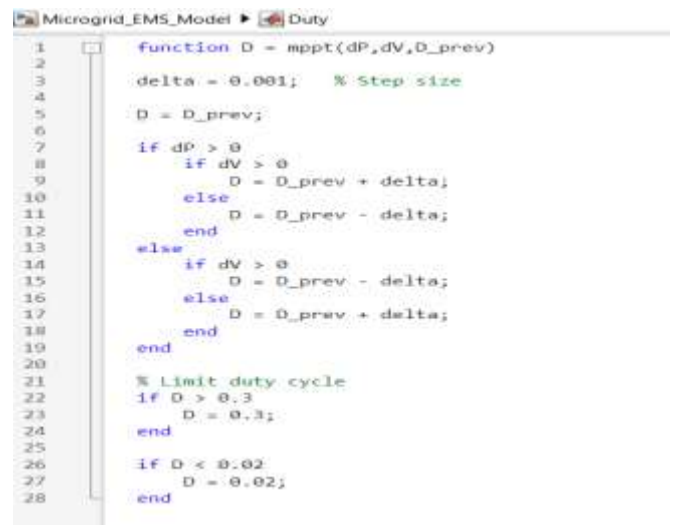
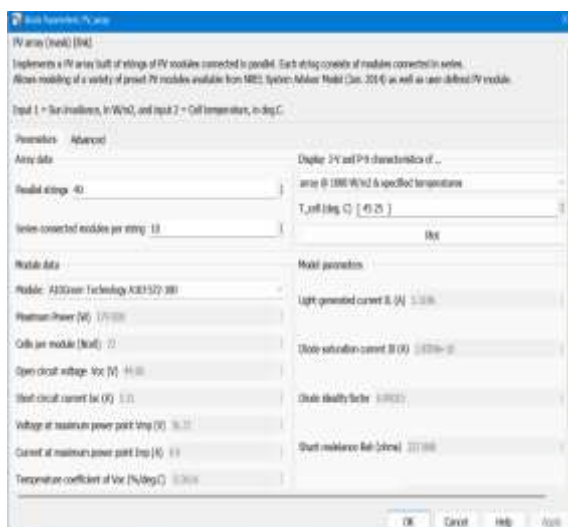
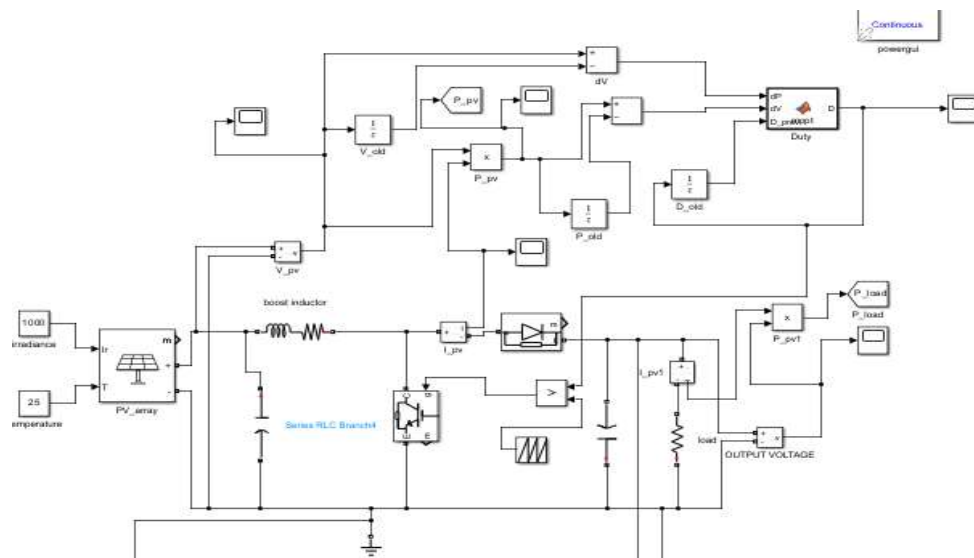
- Step up PV voltage to DC bus level
- Regulate output voltage
- Ensure constant power delivery

### ▪ Working Principle

- Uses switching device (IGBT)
- Stores energy in inductor during ON state
- Releases energy during OFF state

### ▪ Voltage Gain Relation

$$V_{out} = \frac{V_{in}}{1 - D}$$

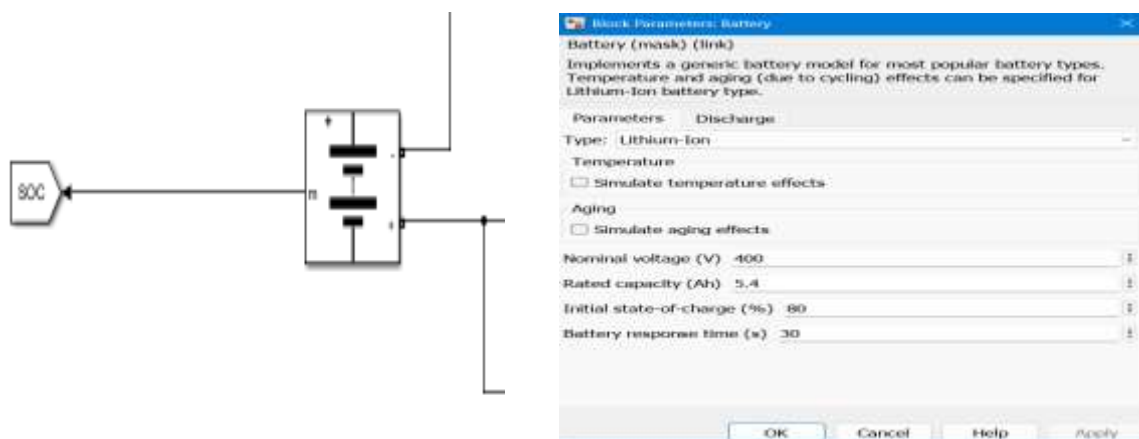


- PV data selection and duty cycle operation

### 3.3 Battery Energy Storage System

- ✓ The battery system is designed with a nominal voltage of 400 V and is connected to the DC bus through a bidirectional DC-DC converter.
- Battery Voltage: 400 V
- SOC operating range: 20% – 90%
- Connected via bidirectional converter

Here, Since the battery voltage 400 V is lower than the DC bus voltage 600 V, direct connection is not possible. Therefore, a bidirectional DC-DC converter is required to match voltage levels and control power flow.



#### ✓ Battery Operation

- Stores excess energy during high PV generation
- Supplies energy when PV generation is insufficient
- Maintains system stability

#### ▪ SOC Constraint

$$SOC_{min} \leq SOC \leq SOC_{max}$$

This is used for,

- Prevent battery damage during overcharging
- reduced lifespan during Deep discharge

### **3.4 Bidirectional DC-DC System**

- ✓ The bidirectional converter interfaces the 400 V battery with the 600 V DC bus, enabling efficient voltage conversion and power flow control.

- Voltage measurements of battery and DC bus

- Battery Side: 400 V
- DC Bus Side: 600 V

Hence we need to step up and step down the voltage accordingly,

- Step-up required -Boost mode
- Step-down required - Buck mode

- **Modes of Operation**

Discharging Mode - Boost Mode

- Power flows: Battery  $\rightarrow$  DC bus
- Voltage conversion: 400 V  $\rightarrow$  600 V
- Used when PV power is insufficient

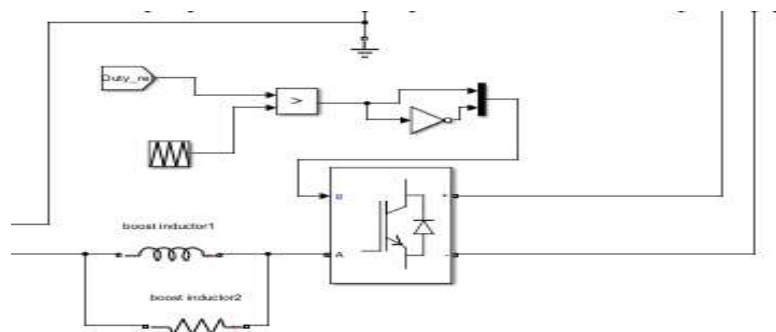
Charging Mode - Buck Mode

- Power flows: DC bus  $\rightarrow$  Battery
- Voltage conversion: 600 V  $\rightarrow$  400 V
- Used when PV power is excess

Here, converter ensures voltage compatibility between subsystems, enables bidirectional energy transfer and improves overall system flexibility and efficiency.

The DC bus is maintained at a constant voltage level of 600 V, which serves as the main power distribution node.

Accordingly, here we were used to 600v DC bus because of provides stable voltage for load operation, allows efficient power transfer, suitable for integration with AC grid.





### **3.5 Energy Management System**

- ✓ The EMS is implemented using a rule-based control algorithm that manages power flow between the PV system, battery, and load.

The net power difference is defined as,

$$P_{diff} = P_{PV} - P_{Load}$$

This parameter determines whether the system operates in charging, discharging, or idle mode.

- ✓ Operating Modes

- **Charging Mode (Excess PV Power)**

**Condition:**  $P_{diff} > 100$

If  $SOC < SOC_{max}$ , the battery is charged  
Converter operates in buck mode (600 V  $\rightarrow$  400 V)  
Duty cycle = 0.8

- **Discharging Mode (Power is below to load power)**

**Condition:**  $P_{diff} < -100$

If  $SOC > SOC_{min}$ , the battery supplies power  
Converter operates in boost mode (400 V  $\rightarrow$  600 V)  
Duty cycle = 0.2

- **Idle Mode (Balanced Condition)**

**Condition:**  $-100 \leq P_{diff} \leq 100$

No battery interaction  
Converter remains inactive

- **Duty Cycle Limitation**

The duty cycle is constrained within:

$$0 \leq D \leq 0.95$$

to ensure stable converter operation and prevent over-modulation.

```

function [mode, duty_ref] = EMS_Controller(P_pv, P_load, SOC)

SOC_max = 90; % Maximum SOC (%)
SOC_min = 20; % Minimum SOC (%)

P_diff = P_pv - P_load;

mode = 0;
duty_ref = 0;

% CASE 1: Excess PV Power → Charge Battery
if P_diff > 100
    if SOC < SOC_max
        mode = -1; % Charging mode
        % Buck mode (500V bus → 400V battery)
        duty_ref = 0.8;
    else
        % Battery full
        mode = 0;
        duty_ref = 0;
    end
end

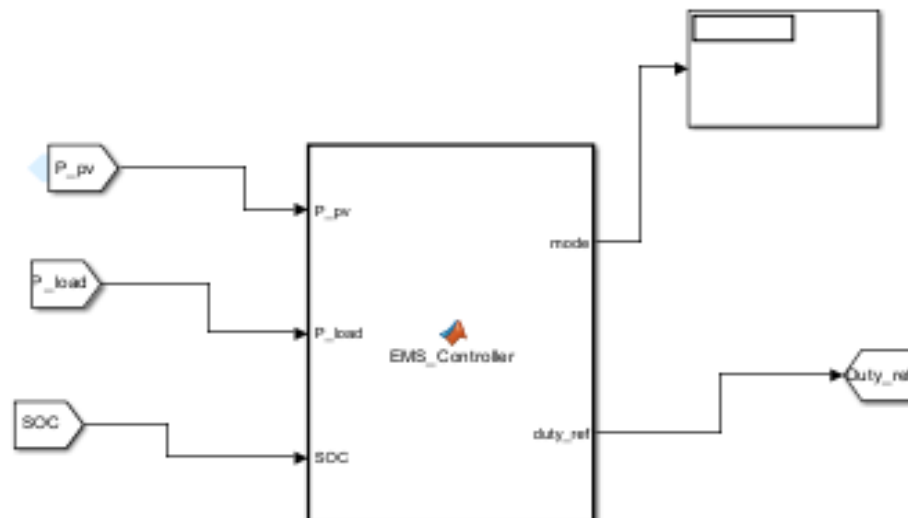
% CASE 2: PV Power Not Enough → Discharge Battery
elseif P_diff < -100
    if SOC > SOC_min
        mode = 1; % Discharge mode
        % Boost mode (400V battery → 500V bus)
        duty_ref = 0.2;
    else
        % Battery empty
        mode = 0;
        duty_ref = 0;
    end
end

% CASE 3: Power Balanced
else
    mode = 0;
    duty_ref = 0;
end

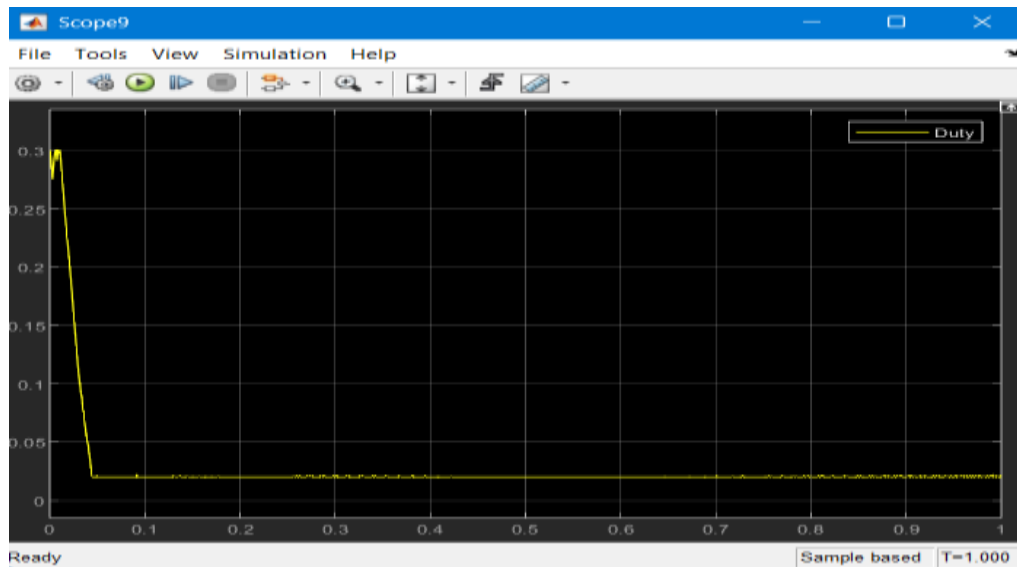
% Safety Saturation area
if duty_ref > 0.95
    duty_ref = 0.95;
end

if duty_ref < 0
    duty_ref = 0;
end
end

```

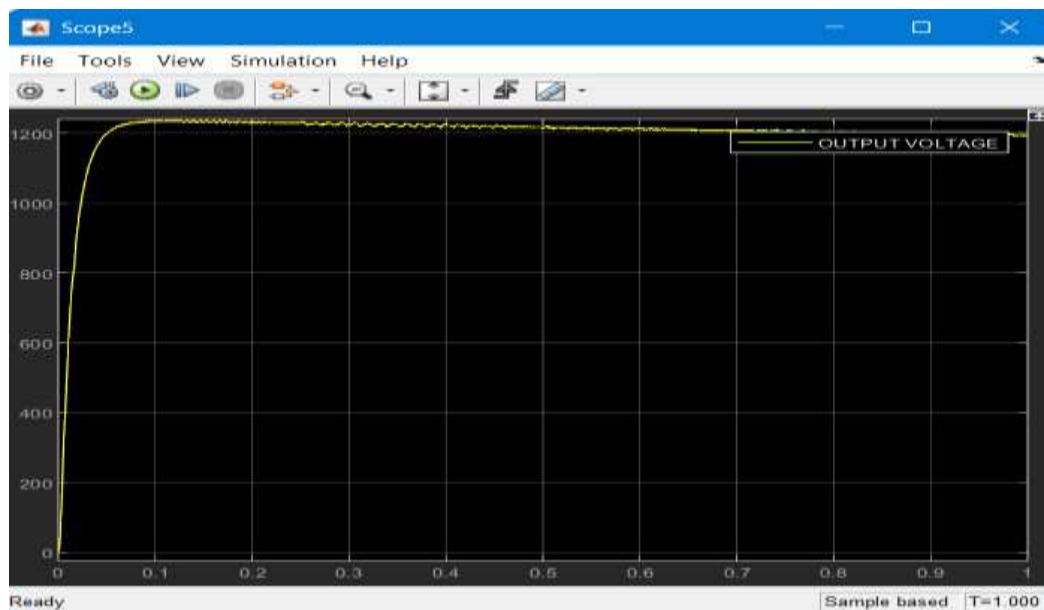


## 4. SIMULATION RESULTS AND ANALYSIS



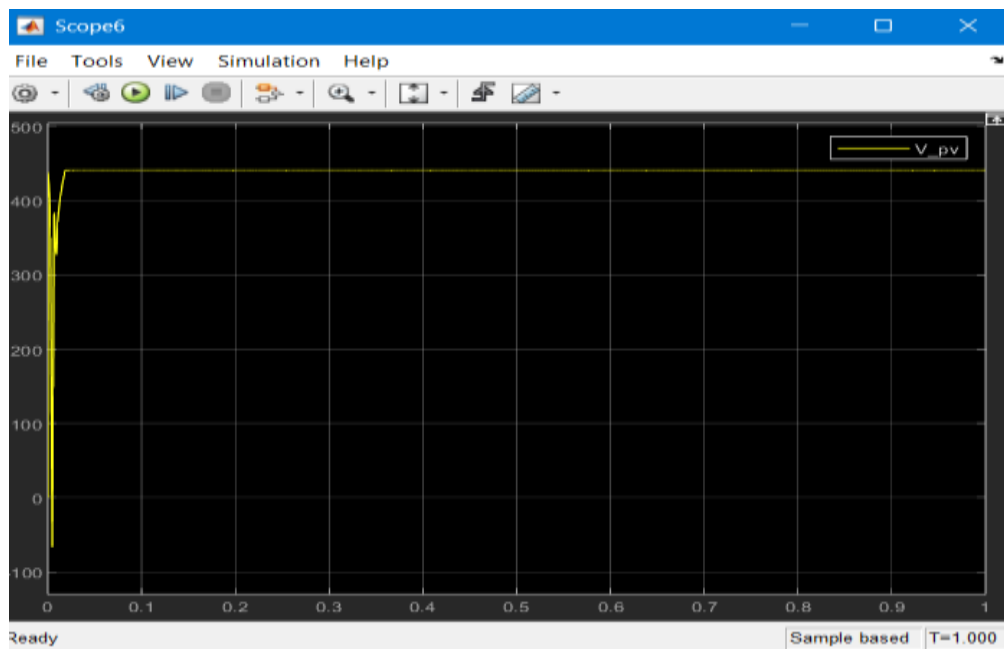
The duty cycle shows a high value at the beginning and quickly decreases to a small steady value.

- Hence, the controller reacts fast at startup, after that, it becomes stable and the converter is properly controlled



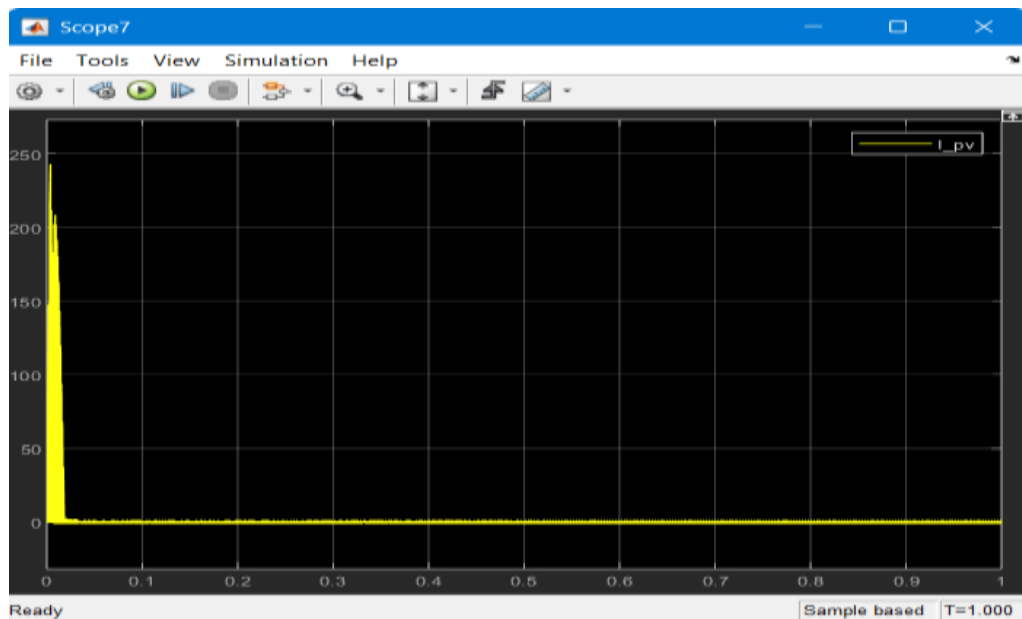
The output (DC bus) voltage rises fast and stabilizes around 1200 V with small ripple.

- Hence, this indicates system can supply the load properly, voltage is stable and only small fluctuations are present.



The PV voltage increases quickly and settles around 450 V.

- Hence, this graph show the boost converter is working correctly, the voltage is properly regulated and the system reaches steady condition quickly.



The PV current shows a large spike at the beginning and then becomes stable.

- This happens because, at startup, there is a sudden energy demand after that, the system reaches normal operation.

## **5. DISCUSSION**

During the Week 13, we did the development and integration of the DC microgrid system. In this work, we connected PV system, boost converter, battery system, bidirectional converter and EMS together.

At the beginning stage, some errors were occurred in different parts of the system. In the PV system, simulation error was coming because of the nonlinear behaviour of the PV diode. Sometimes simulation stopped and showed instability. to fix this, we changed solver settings and added small resistance to make the system stable.

In the boost converter section, high voltage and current spikes were appeared at the starting time. This was mainly due to improper initial values and sudden duty cycle changes. then we corrected the initial conditions and improved the duty cycle control, so the system became stable. In the battery system, charging and discharging was not working correctly at first. this happened because SOC limits were not properly defined. after that, we added SOC minimum and maximum limits, so battery works properly.

In the EMS logic, sudden switching between modes caused instability. The duty cycle was changing very fast. to solve this, we added threshold limits and improved the logic, so mode changes are smooth. after making these corrections, the system performance improved well. the PV voltage becomes stable, output voltage is regulated, and battery charging and discharging works correctly. EMS also controls the system properly.

From the simulation results, we can see that after small initial transients, the system reaches steady state condition and works in stable manner. Finally, we can say that the errors in the previous design were corrected and the system is now working properly with good energy management.

## **6. CONCLUSION**

In Week 13, we did the development of the DC microgrid system. In this system, we used PV, boost converter, battery, bidirectional converter and EMS.

At first, some errors were coming in the system. PV section had instability problem. boost converter had high spikes at starting. battery was not working correctly. EMS also had sudden switching problem. after that, we fixed those problems. we changed solver settings, added small resistance, gave proper initial values and improved EMS logic. also we added SOC limits for battery.

After doing these changes, the system is working properly. PV voltage is stable. Output voltage is also stable. Battery is charging and discharging correctly. EMS is controlling the system well. from the results, we can see that after starting time, the system becomes stable and works normally.

Finally, we can say that the system is working correctly and energy management is done properly. This system can be used for next stage grid connection.

## **7. REFERENCES**

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